SHI FENG

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EDUCATION

Tsinghua University *Sep 2019 - Jul 2023 (expected)*
Bachelor of Science, selected to Yao Class
• GPA: 3.87/4.0
• Selected Courses: Introduction to AI (A), Mathematics for Computer Science (A), Causal and Statistical Learning (A+), Abstract Algebra (A), Introduction to Databases (A+), Artificial Intelligence: Principles and Techniques (A), Theory of Computation (A), Distributed and Blockchain System (A-), Machine Learning (A), Operating System (A), Research Immersion Training (A+)

RESEARCH INTERESTS

Theoretical computer science and economics, particularly algorithmic game theory, learning theory, causality, and machine learning.

RESEARCH EXPERIENCES

Multi-Agent Owner-Assisted Scoring Mechanisms *Aug 2022 - present*
Advisors: Prof. **Yuhao Wang**, Prof. **Weijie Su** Tsinghua IIIS&UPenn Wharton School
• Extended owner-assisted scoring problem to a multi-agent setting and wrote a research proposal for it.
• Designed a fair, "exchangeable noise assumption free", and truthful mechanism for the extended problem.

Peer Prediction for Learning Agents *Feb 2022 - Jun 2022*
Advisors: Prof. **Yiling Chen**, Prof. **Fang-Yi Yu** Harvard EconCS Group (in-person)
• Surveyed mechanisms for information elicitation and extended the original multi-task setting to a sequential setting.
• Proved truthful convergence of correlated agreement mechanism adopted on agents using reward-based online learning algorithms and a negative result showing no-regret assumption on agents can-not guarantee truthful convergence.
• Created a peer prediction simulator and conducted numerical experiments which supported theoretical results.

Combinatorial Causal Bandits *Jul 2021 - Feb 2022*
Advisor: Prof. **Wei Chen** MSR Asia Theory Center
• Surveyed causal bandits algorithms for pure exploration and cumulative regret minimization.
• Presented the new combinatorial causal bandit framework with a definition of cumulative regret.
• Proposed algorithms for combinatorial causal bandits problem on binary generalized linear models with hidden variables.
• Implemented algorithms in OOP and conducted numerical experiments to show their efficiency.

Causal Inference for Social Networks *Nov 2020 - Jul 2021*
Advisor: Prof. Wei Chen MSR Asia Theory Center
• Surveyed identifiability of Bayesian causal models and completeness of the do-calculus algorithm.
• Provided transformations from an Independent Cascade Model to a Bayesian causal model.
• Proved necessary and sufficient conditions for identifiability in three common types of Independent Cascade Models.

PUBLICATIONS

(* denotes equal contribution)

Shi Feng, Wei Chen. Combinatorial Causal Bandits.
The 37th AAAI Conference on Artificial Intelligence (AAAI 2023).
[arXiv] [code]

Shi Feng, Fang-Yi Yu, Yiling Chen. Peer Prediction for Learning Agents.
The 36th Conference on Neural Information Processing Systems (NeurIPS 2022).
[arXiv] [code]

Shi Feng, Wei Chen. Causal Inference for Influence Propagation–Identifiability of the Independent Cascade Model.
The 10th International Conference on Computational Data and Social Networks (CSoNet 2021) **Best Paper**.
[arXiv] [publication]

Shi Feng*, Zimeng Song*, Weijie Su, Yuhao Wang. A Truthful Multi-Agent Owner-Assisted Scoring Mechanism.
Writing in progress.
[slides]

SELECTED HONORS & AWARDS

Comprehensive Excellence Awards *(top 10%)* *Oct 2020, 2022*, THU
"Star of Tomorrow" Award of Excellence *Mar 2022*, MSR Asia
Best Paper Award *(1/57)* *Nov 2021*, CSoNet
Innovation Excellence Award *Oct 2021*, THU

Social Practice Excellence Award *Oct 2021*, THU
"Challenge Cup" Science and Technology Competition, Top Grade Prize *(top 3%)* *Apr 2021*, THU
Freshman Scholarship *(top 10%)* *Oct 2019*, THU
Russian Mathematical Olympiad, Silver Medal *Apr 2019*, MSE of Russia
Chinese Mathematical Olympiad, Gold Medal *Nov 2018*, CMS
National High School Mathematics League, First Prizes *Sep 2016, 2017, 2018*, CMS

TALKS

NeurIPS 2022, Peer Prediction for Learning Agents *Oct 2022*
Yao Class Seminar, Peer Prediction for Learning Agents *Sept 2022*
CSoNet 2021, Causal Inference for Influence Propagation–Identifiability of the Independent Cascade Model *Nov 2021*

SERVICES

Class Leader, Yao Class 91, Tsinghua University *Aug 2022 - present*
Reviewer, Journal of Combinatorial Optimization (JOCO) *May 2022*
Team Member, Chinese national team for Russian Mathematical Olympiad *Apr 2019*

SKILLS

Programming Languages: C/C++, Python, Golang, SQL, assembly language, Bash, HTML
Professional Applications: Latex, Mathematica, Matlab, PyTorch, TensorFlow, Git
Languages: Chinese (native language), English (fluent)
Hobbies: soccer (member of IIIS soccer team), road trip (driving through China National Highway 318 in 2019), handball (amateur training for one year), basketball, vehicle engineering (1 invention patent and 1 utility model patent), Go (amateur 1 dan)